

2. Low-Level Document (LLD) (Expanded)

Introduction:

The **Low-Level Document (LLD)** provides a deep dive into the technical details of the project. It complements the HLD by explaining the specific steps and components used in the Power BI dashboard, focusing on data transformation, DAX calculations, and how the dashboard elements work together.

Scope:

The scope of the LLD is to provide a technical breakdown of the architecture, data processing, and implementations used in creating the Power BI dashboard. This includes the data flow, relationship between tables, and DAX calculations used for KPIs. It also covers the interactivity features (like slicers) and visualization techniques employed to present the data effectively.

Architecture:

1. Power BI Architecture:

- **Power BI Desktop:** This tool is used for data ingestion, transformation, modeling, and visualization creation. It allows the creation of detailed reports and dashboards.
- **Power BI Service:** Once the dashboard is created in Power BI Desktop, it is published to the Power BI Service for sharing and collaboration.
- **Power BI Gateway:** If data is stored on-premises, Power BI Gateway is used to connect to the on-premises data sources and schedule data refreshes.

2. Power BI Service Architecture:

- **Sharing and Collaboration:** Reports and dashboards are shared with authorized users in the organization. The service also allows users to view and interact with the reports on web and mobile.
- **Scheduled Data Refresh:** Ensures that the data remains current by setting up automatic refreshes of data sources.

3. Data Server:

- Data is pulled from external sources, such as SQL databases or Excel files, and then loaded into Power BI for analysis. These data sources might be hosted in the cloud (e.g., AWS, Azure) or on-premises.
 - Data is cleaned, transformed, and modeled in Power BI, ready to be visualized.
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Architecture Description:

The architecture begins with the extraction of raw data from external sources, which is loaded into Power BI Desktop. In the desktop environment, Power Query is used to clean and transform the data (e.g., converting date formats, handling missing values). Once the data is prepared, the relationship between tables is defined, and DAX measures are written to calculate KPIs. After visualizations are created, the dashboard is published to Power BI Service, where it can be shared, viewed, and refreshed automatically.

Unit Test Cases:

- **Test Case 1: Validate Revenue Calculation**
 - **Input:** A set of sales records from multiple regions (e.g., North America, Europe, Asia).
 - **Expected Output:** Total revenue calculated should match the aggregate sales revenue for each region.
 - **Test Case 2: Verify Filter Application**
 - **Input:** Apply filters by region and year.
 - **Expected Output:** The dashboard should update to show only the sales data for the selected region and year.
 - **Test Case 3: Test Date Format Transformation**
 - **Input:** A dataset with multiple date formats.
 - **Expected Output:** All dates should be transformed into a consistent format (e.g., DD/MM/YYYY) using the DAX function FORMAT.
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The above sections can be expanded further by continuing to provide additional details in subsequent sections, such as **Wireframe**, **Project Code**, **Clustering**, and **Detailed Project Report**. This expansion adds complexity and depth to each part, ensuring that your documentation is robust and easy to follow for both technical and non-technical audiences.